

Catch-Up without Convergence: Sectoral TFP Dynamics and Persistent Regional Heterogeneity in Northeastern Brazil

Abstract. Northeastern Brazil exhibits short-run productivity catch-up alongside widening differences in cumulative TFP paths. Using an annual state-sector panel, we estimate total factor productivity for the nine states from 1986 to 2023 in agriculture, industry, commerce and services, and the aggregate economy, allowing production elasticities to evolve over time. From 1995 onward, lower relative TFP predicts faster subsequent growth in every sector, yet cross-state dispersion rises throughout the economy. Agriculture combines the largest productivity gains with the sharpest polarization, separating states into persistent convergence clubs. In manufacturing, dispersion increases in all seven industries observed continuously across the nine states, showing that sectoral composition alone cannot account for industrial divergence. Five-year transitions confirm persistent low- and high-productivity positions. Within-sector productivity dynamics thus help explain why temporary catch-up does not translate into regional convergence.

Keywords: Total factor productivity; Regional divergence; Structural transformation.

JEL Classification: O47; R11; R12.

Catch-up sem convergência: dinâmica setorial da PTF e persistência da heterogeneidade regional no Nordeste brasileiro

Resumo. O Nordeste brasileiro apresenta catch-up de produtividade no curto prazo ao mesmo tempo que as trajetórias cumulativas de PTF se tornam mais heterogêneas. Usando um painel anual estadual-setorial, estimamos a produtividade total dos fatores dos nove estados entre 1986 e 2023 na agropecuária, indústria, comércio e serviços e economia agregada, permitindo que as elasticidades de produção variem no tempo. Desde 1995, menor PTF relativa prediz maior crescimento subsequente em todos os setores, enquanto a dispersão interestadual aumenta. Na agropecuária, os maiores ganhos de produtividade coexistem com forte polarização e clubes persistentes. Na indústria, a dispersão aumenta nas sete atividades observadas continuamente nos nove estados, mostrando que a composição setorial não explica sozinha a divergência industrial. Transições de cinco anos confirmam a persistência das posições extremas. As dinâmicas de produtividade intrassetoriais ajudam, assim, a explicar por que o catch-up temporário não se traduz em convergência regional.

Palavras-chave: Produtividade total dos fatores; Divergência regional; Transformação estrutural.

1. Introduction

Within Northeastern Brazilian states, lower lagged relative productivity predicts faster subsequent growth, yet the regional distribution becomes more dispersed. Between 1995 and 2023, the cross-state standard deviation of log total factor productivity (TFP) rises from 0.089 to 0.234 in the aggregate economy, from 0.640 to 1.321 in agriculture, from 0.144 to 0.494 in industry, and from 0.188 to 0.625 in commerce and services. Over the same period, panel estimates produce a negative coefficient on lagged relative TFP in every sector. The two patterns describe different margins of regional dynamics: short-run correction in growth rates can coexist with widening gaps in cumulative productivity paths, as distributional approaches to convergence emphasize (Quah, 1993, 1997). In the Northeast, catch-up occurs without long-run compression of the productivity distribution.

This distinction matters for regional development because faster growth among lagging places is economically different from broad diffusion of productive efficiency. Investment and sectoral expansion can raise output while leaving relative productivity positions largely intact when efficiency gains remain concentrated. Northeastern Brazil offers a particularly informative setting for this question. Its nine states share national institutions and broad macroeconomic conditions, yet combine different agricultural structures, industrial bases, labor-force profiles, and paths of structural transformation.

The relevant convergence margin is therefore whether productivity improvements within sectors spread widely enough to alter the regional hierarchy.

We study that margin with an annual state-sector panel spanning 1986–2023. TFP is recovered for agriculture, industry, commerce and services, and the aggregate economy using production measures tailored to each sector. Physical capital stocks are constructed through the perpetual-inventory method where a coherent investment series is available; agriculture additionally includes land, while electricity per worker measures operating capacity in commerce and services. Almost four decades encompass macroeconomic stabilization, agricultural restructuring, industrial restructuring, and substantial changes in production. We therefore allow input elasticities to evolve smoothly over time using Flexible Least Squares (FLS) (Kalaba and Tesfatsion, 1988, 1989), while anchoring their long-run means to fixed-coefficient OLS estimates. Each TFP index is normalized to its own state-sector value in 1986, so the empirical object is the cumulative productivity path relative to a common within-state benchmark.

Three pieces of evidence establish the central result. Cross-state dispersion rises in all four sectors even when the convergence window begins in 1995, after the common index normalization has ceased to constrain the cross section. At the same time, negative beta coefficients under state and year fixed effects and wild-cluster inference reveal systematic short-run correction. Medium-run transitions show why that correction fails to close the distribution: five years later, 87.9% of agricultural observations initially in the low-productivity regime remain low and 86.2% of observations at the top remain high; the corresponding probabilities in industry are 80.9% and 74.7%. Faster growth after falling behind is therefore common, while durable movement across the distribution remains limited.

Sectoral evidence explains how that combination is sustained. Agriculture records the strongest productivity expansion, but those gains are sharply uneven across space. The Phillips–Sul procedure identifies three transition clubs: Alagoas, Bahia, Paraíba, and Sergipe form the leading group; Ceará, Pernambuco, and Rio Grande do Norte occupy an intermediate path; and Maranhão and Piauí follow the weakest cumulative trajectories. Average TFP in the leading club rises from 79.1 in 1995 to 287.9 in 2023, while the lagging club moves from 23.0 to 11.9. Agricultural dispersion continues to widen after Bahia, Maranhão, and Piauí are excluded jointly, extending the pattern beyond the states associated with the MATOPIBA frontier. Industry reveals a different source of persistence. Cross-state TFP dispersion increases in all seven manufacturing activities observed continuously in all nine states, and average within-industry dispersion rises by roughly 65% between 2005–2009 and 2019–2023. Industrial dispersion therefore widens even after broad activity is held fixed.

The paper advances two related literatures. Research on Brazilian regional productivity documents periods of state-level catch-up and assigns an important role to within-sector growth, particularly in manufacturing and services (Canêdo-Pinheiro and Barbosa Filho, 2011; Schettini and Azzoni, 2018). Work focused on the Northeast also shows that spatial dependence and access to common productive factors shape municipal productivity growth (Anderle et al., 2018). We extend this evidence by following annual state-sector TFP over a longer horizon and separating short-run corrective growth from changes in the regional distribution. A second literature asks whether regional productivity gaps reflect sectoral composition or differences in the efficiency of similar activities across places. The sectoral evidence of Bernard and Jones (1996) shows that aggregate convergence can conceal heterogeneous industry dynamics, while recent firm-level evidence for lagging European regions finds that much of the productivity penalty persists within sectors (Menon and Vermeulen, 2026). Our balanced manufacturing comparison brings that distinction to a lagging region of a large emerging economy and shows widening dispersion inside every continuously observed activity.

The broader implication is that structural transformation can make a lagging macro-region more productive without making it internally more homogeneous. Agricultural productivity upgrading can generate large efficiency gains while concentrating them in a subset of places, and comparable manufacturing activities can follow increasingly different productivity paths across states. Regional convergence should therefore be evaluated by the diffusion of productivity gains across the distribution, not solely by episodes of faster growth among places that have fallen behind. Section 2 positions this

argument in the convergence and regional-productivity literatures. Section 3 presents the data, TFP measurement, and distributional framework; Section 4 develops the sectoral evidence; and Section 5 concludes.

2. Related Literature and the Regional Productivity Puzzle

Convergence has two empirically distinct dimensions. In the conventional growth-regression approach, a negative relationship between an initial productivity level and subsequent growth indicates that relatively lagging places tend to expand faster (Barro and Sala-i-Martin, 1992). Distributional approaches instead ask whether cross-sectional gaps narrow and whether economies move persistently through the relative distribution. Quah (1993) showed why regression toward the mean can coexist with a dispersed distribution, and later work formalized stratification, polarization, and convergence clubs as distinct transition outcomes (Quah, 1997; Phillips and Sul, 2007, 2009). This distinction organizes our empirical design: beta estimates measure corrective growth, while sigma dispersion, convergence clubs, and medium-run transitions reveal whether that correction alters the regional hierarchy.

Brazilian evidence makes the distinction particularly relevant. Using labor productivity for Brazilian states, Canêdo-Pinheiro and Barbosa Filho (2011) find weak convergence during 1985–1995 and a faster process after 1995, with within-sector productivity growth playing the predominant role and manufacturing and services contributing importantly to the aggregate pattern. Employing stochastic frontiers for agriculture, industry, services, and total production, Schettini and Azzoni (2018) identify strong agricultural efficiency growth together with signs of state convergence, especially in industry. Sectoral TFP estimates at the macro-regional level also reveal substantial variation in the sources of Brazilian productivity performance (Souza and Cunha, 2018; Santos and Spolador, 2018). Those findings establish the importance of the sectoral margin. Our question concerns the longer-run distributional counterpart: whether faster growth among lagging states compresses productivity gaps within a historically lagging macro-region.

Evidence centered on Northeastern Brazil also points to an explicitly regional process. Anderle et al. (2018), in this journal, document spatial dependence in municipal productivity growth and patterns consistent with neighboring municipalities sharing and competing for scarce productive factors. Brazilian manufacturing location is highly localized, and the spatial reach of agglomeration forces decays rapidly with distance (Almeida et al., 2023). Recent work on regional diversification further shows that entry into new activities is strongly conditioned by capabilities related to the existing productive structure (Françoso et al., 2024; Freitas et al., 2024). These findings give economic content to persistent within-sector gaps: the same broad activity can operate within very different local productive environments.

The distinction between sectoral composition and within-sector productivity is equally important internationally. Bernard and Jones (1996) demonstrate that aggregate convergence need not be reproduced uniformly across sectors, making disaggregation essential for understanding productivity dynamics. More recently, Menon and Vermeulen (2026) use firm-level data for Italy and Spain and find that sectoral composition explains only a limited share of the productivity gap between less and more developed regions; most of the difference remains as a multifactor-productivity penalty within sectors. Our industrial exercise applies the same economic distinction at a different spatial scale. Seven manufacturing activities observed continuously in all nine Northeastern states allow a direct test of whether regional productivity paths continue to separate when broad activity is held constant.

Agriculture adds a second dimension to the puzzle. Structural change in Brazilian agriculture has generated substantial productivity gains, while recent evidence emphasizes how unevenly productivity gains and structural change are distributed across space. Agricultural economic complexity is closely associated with regional productivity and development outcomes (Stein et al., 2024); within the Northeast, Antunes and Freitas (2025) document marked productivity heterogeneity across microregions and distinguish the MATOPIBA frontier from the rest of the region. Our state-level TFP paths capture cumulative efficiency growth for entire state agricultural sectors. They reveal a clear internal stratification of the Northeast, and the widening persists after the Northeastern MATOPIBA

states are removed jointly.

Against this background, our contribution is to connect state catch-up, distributional persistence, and within-sector heterogeneity in a single long state-sector panel. Short-run correction and long-run divergence are jointly organized within the same lagging regional system: agricultural productivity upgrading creates persistent productivity clubs, manufacturing productivity paths separate inside comparable activities, and medium-run mobility remains limited. The evidence therefore shifts attention from whether lagging places grow faster to whether productivity upgrading diffuses widely enough to change the regional distribution.

3. Data and Empirical Framework

3.1. Data and sector-specific production measures

We assemble an annual panel for the nine states of Northeastern Brazil over 1986–2023. State GDP and sectoral value added come from the Regional Accounts produced by the Brazilian Institute of Geography and Statistics (IBGE), accessed through Ipeadata. Employment and schooling are constructed from the Annual Social Information Report (Relação Anual de Informações Sociais, RAIS), so the labor denominator follows a consistent formal-employment concept across states and years; average years of schooling measures observable labor quality within that same workforce. Table 1 summarizes the production measures. A common growth-accounting logic is maintained across sectors while productive capacity is represented by inputs appropriate to each activity.

Physical capital is constructed through the perpetual-inventory method (PIM),

$$K_{it} = (1 - \delta)K_{i,t-1} + I_{it}, \quad (1)$$

where I_{it} denotes investment and δ the depreciation rate. For the aggregate economy, national machinery and equipment investment at constant 2010 prices is regionalized using state GDP shares. The industrial series applies the same national investment aggregate and allocates it using state industrial value-added shares, with the Northeast component linked to the national benchmark in 2010. Agricultural investment is proxied by state rural-credit operations contracted for investment. We use depreciation rates of 10% for the aggregate economy, 13% for industry, and 9% for agriculture, following the capital-stock calibration in Souza Júnior and Cornelio (2020). The resulting stocks are divided by the corresponding worker counts before entering the production equations.

Agricultural land is the planted or harvest-destined area reported by IBGE’s Municipal Agricultural Production survey (*Produção Agrícola Municipal*). The series is observed from 1988 onward; values for 1985–1987 required to initialize the growth series are extrapolated from the initial linear trend. Land enters the agricultural growth equation in levels as the sector-specific fixed factor, while output and physical capital are expressed per worker. Commerce and services use electricity consumption per worker from the Ministério de Minas e Energia as an operating-capacity measure. State-level electricity consumption is unavailable in 1997, leaving 37 annual observations per state in that sector. RAIS schooling provides a consistent long-run measure of observable labor quality among formal workers.

Table 1. Sector-specific production measures and data sources

Sector	Output measure	Productive capacity and fixed inputs	Human capital
Aggregate economy	State GDP per worker	Physical-capital stock per worker, built by PIM from national machinery and equipment investment and regionalized by state GDP shares	Average years of schooling among formal workers
Industry	Industrial value added per worker	Physical-capital stock per worker, built by PIM and regionalized using industrial value-added shares	Average years of schooling among formal industrial workers
Agriculture	Agricultural value added per worker	Physical-capital stock per worker, built by PIM from rural investment credit; planted or harvest-destined area enters as a sector-specific fixed factor	Average years of schooling among formal agricultural workers
Commerce and services	Value added per worker	Electricity consumption per worker as an operating-capacity measure	Average years of schooling among formal workers in commerce and services

Notes: PIM denotes the perpetual-inventory method. Monetary output series are expressed at constant 2010 prices. The commerce-and-services panel has no state-level electricity observation in 1997.

Source: Authors' construction from IBGE Regional Accounts/Ipeadata, RAIS, Banco Central do Brasil rural-credit data, IBGE's Produção Agrícola Municipal, and Ministério de Minas e Energia electricity series.

A harmonized manufacturing panel for 1998–2023 provides a second level of comparison at the state–industry–year level. Real value of industrial transformation and productive capital are built from IBGE's Annual Industrial Survey (*Pesquisa Industrial Anual*, PIA), while employment is adjusted for RAIS schooling to form effective labor; monetary variables are expressed in 2023 prices. Cell-level TFP is estimated from real output, real capital, and effective labor using the same OLS-anchored FLS logic described below. Seven broad manufacturing activities are continuously observed in every state over the full window: apparel; wood products; printing; chemicals and pharmaceuticals; rubber and plastics; non-metallic minerals; and furniture, miscellaneous manufacturing, and repair. The balanced sample contains 63 state–industry cells and lets us compare productivity dispersion across locations within the same broad activity.

3.2. TFP estimation with time-varying technology

A fixed production relationship would keep the contribution of productive inputs unchanged from the late 1980s to the early 2020s. Over a period that spans macroeconomic stabilization, agricultural restructuring, and industrial restructuring, we allow those contributions to move gradually over time. Let y_{ist} denote output per worker for state i , sector s , and year t , and let x_{mist} collect the relevant productive inputs. The growth-accounting representation is

$$\Delta \ln y_{ist} = \alpha_{ist} + \sum_m \beta_{mist} \Delta \ln x_{mist} + \varepsilon_{ist}, \quad (2)$$

where aggregate and industrial specifications include physical capital per worker and human capital, agriculture additionally includes land, and commerce and services combine the electricity-based operating-capacity measure with human capital. TFP growth is the portion of output growth left after the estimated contribution of observed inputs,

$$\Delta \ln A_{ist} = \Delta \ln y_{ist} - \sum_m \widehat{\beta}_{mist} \Delta \ln x_{mist}. \quad (3)$$

With an intercept in Equation (2), this component equals the estimated intercept plus the regression residual.

Flexible Least Squares (FLS) chooses coefficient paths that balance annual fit with temporal smoothness (Kalaba and Tesfatsion, 1988, 1989). For each of the 36 state–sector series, the estimator minimizes

$$\sum_{t=1}^T (\Delta \ln y_t - \mathbf{z}_t' \boldsymbol{\beta}_t)^2 + \mu \sum_{t=2}^T (\boldsymbol{\beta}_t - \boldsymbol{\beta}_{t-1})' (\boldsymbol{\beta}_t - \boldsymbol{\beta}_{t-1}), \quad (4)$$

where $\mathbf{z}_t$ contains a constant and the relevant input growth rates. The first term rewards fit and the second rewards smooth coefficient paths; larger values of μ place greater weight on temporal stability.

For each state–sector model, μ is selected on a logarithmic grid from 10^1 to 10^5 . The criterion maximizes the gain in fit relative to fixed-coefficient OLS subject to a long-run anchor: the time average of every FLS coefficient must lie within one Newey–West OLS standard error of its corresponding OLS estimate. The procedure therefore permits gradual technological adjustment while retaining the long-run information in the sample. Appendix Table A1 reports the convergence diagnostics under fixed-coefficient OLS, alternative anchoring bands, and an FLS specification that constrains input coefficients to be nonnegative.

Annual TFP indices are obtained by recursively cumulating Equation (3) and normalizing each state–sector path to 100 in 1986. This normalization fixes a common origin for cumulative change: a value of 200 indicates that the estimated TFP index doubled relative to that state’s own 1986 benchmark. Accordingly, cross-state comparisons below concern the dispersion, transition, and persistence of cumulative productivity paths, while the index height is not an absolute cross-state ranking of TFP levels.

3.3. Convergence, distributional dynamics, and within-industry comparisons

The empirical analysis separates three questions. First, do cross-state productivity distances narrow? For sector s in year t , sigma dispersion is

$$\sigma_{st} = \text{sd}_i[\ln(A_{ist})]. \quad (5)$$

All state–sector indices equal 100 in 1986, so the base-year cross section has zero dispersion by construction. The main convergence window begins in 1995, when the distribution is already non-degenerate and the Brazilian economy has entered the post-stabilization period. Appendix Table A2 reports windows beginning in 1987 and 2000 as additional comparisons.

Second, do states that fall behind grow faster subsequently? We estimate

$$\Delta \ln A_{ist} = \beta_s \ln A_{is,t-1} + \eta_i + \lambda_t + u_{ist}, \quad (6)$$

separately by sector, with state and year fixed effects. A negative β_s measures short-run corrective growth: lower lagged relative TFP predicts faster TFP growth in the following year. With nine state clusters, inference is based on a state-level wild-cluster bootstrap with Rademacher weights. We interpret the beta coefficient jointly with the evolution of sigma because the two statistics address different margins of convergence.

Third, do states move persistently through the relative productivity distribution? The Phillips–Sul

log- t procedure tests whether relative TFP paths approach a common transition path and, after rejection of global convergence, applies its clustering algorithm to identify groups with statistically compatible transitions (Phillips and Sul, 2007, 2009). Mobility is then summarized with Markov transitions. For each sector, annual relative log TFP is defined around the corresponding cross-state mean, and low, middle, and high regimes are formed from the pooled sector-specific terciles of these relative observations. Five-year transitions capture medium-run movement across the distribution. Appendix Table A3 reports the full club classification and additional transition statistics.

The balanced manufacturing panel sharpens the industrial comparison by holding broad activity constant. Cell TFP paths share a common 1998 normalization, so the baseline comparison begins only after the distribution has had several years to develop. For each of the seven activities observed continuously in all nine states, we compute the annual cross-state standard deviation of log cell-level TFP and compare mean dispersion in 2005–2009 with 2019–2023. We also calculate five-year regime transitions across the 63 state–industry cells. Later early windows, including 2008–2012, and fixed-coefficient cell TFP estimates are reported in Appendix Table A4; every activity continues to widen under those comparisons. This design distinguishes differences in industrial composition from productivity paths that separate across locations within comparable activities.

4. Results and Discussion

4.1. Sectoral productivity paths

No single productivity trajectory characterizes Northeastern Brazil. Figure 1 shows a comparatively smooth aggregate index alongside much wider separation in agriculture, industry, and commerce and services. Since every series starts from its own 1986 value, the vertical scale records cumulative TFP change within each state rather than absolute productivity levels across states. By 2023, aggregate paths remain relatively compressed, while the sectoral panels reveal much larger differences in cumulative productivity change.

Agriculture displays the broadest range of cumulative TFP movements. Alagoas, Paraíba, Sergipe, and Bahia finish far above their starting values, whereas Maranhão and Piauí follow sharply weaker paths. Industry exhibits a different pattern: Bahia is the only state ending clearly above its 1986 industrial benchmark, while the remaining states finish below 100, several by a wide margin. Commerce and services occupy an intermediate position, with sizeable gains in some states alongside growing dispersion. Aggregation therefore smooths sector-specific dynamics that are central to the regional distribution of productivity.

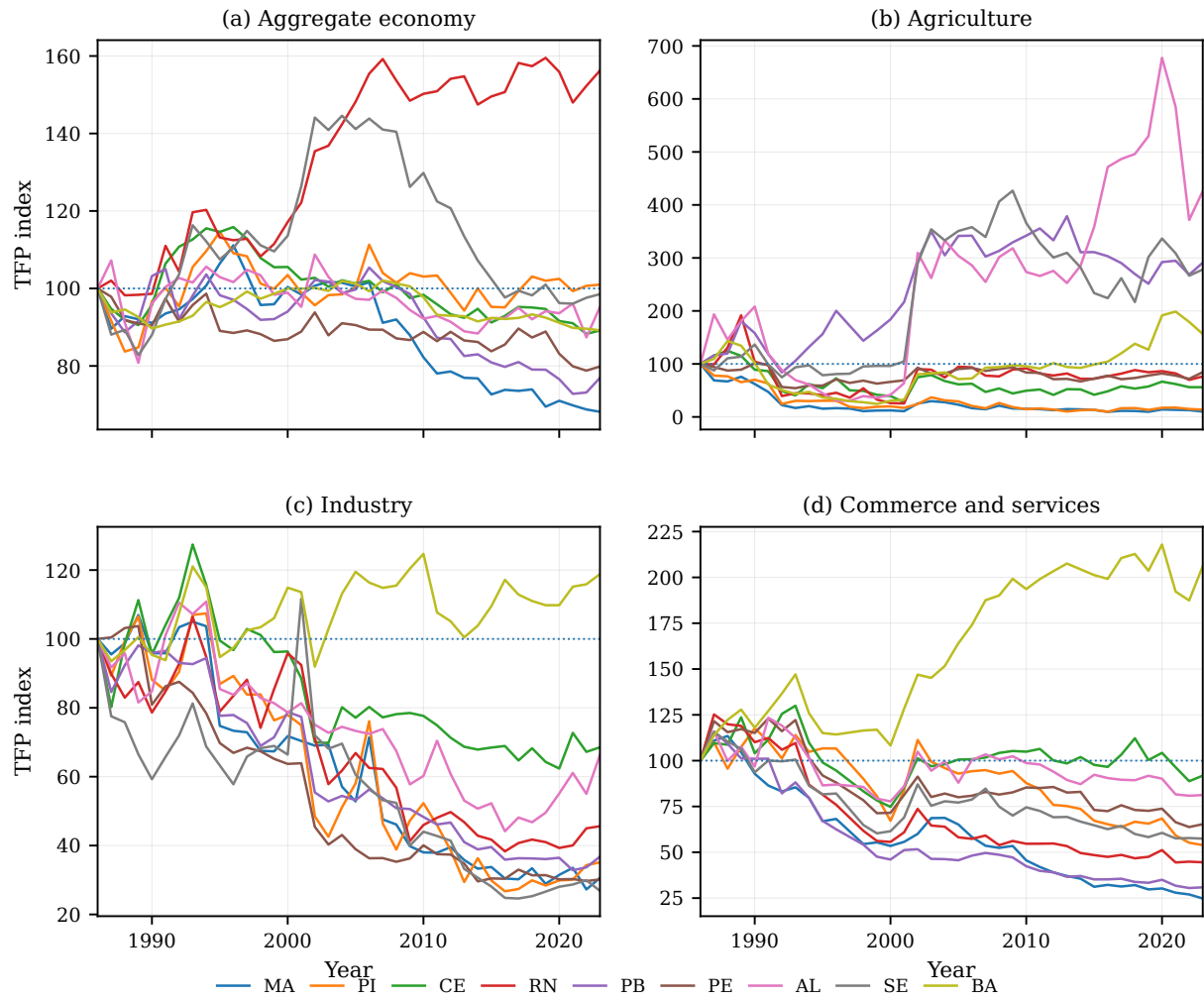
The sectoral contrast already indicates why an aggregate convergence statistic is incomplete. Agricultural TFP can expand rapidly while its gains remain concentrated in a subset of states, whereas weak industrial performance can coexist with growing differences among locations. Commerce and services adds a third pattern of increasing dispersion. The next subsection asks whether faster subsequent growth among relatively lagging observations is strong enough to compress any of these distributions.

4.2. Catch-up without convergence

Table 2 brings the paper’s central convergence evidence together. Panel A reports a negative coefficient on lagged relative TFP in every sector. With state and year fixed effects, $\hat{\beta}$ equals -0.070 for the aggregate economy, -0.115 for agriculture, -0.177 for industry, and -0.065 for commerce and services. Wild-cluster p -values range from 0.006 to 0.010. States that move to relatively low TFP positions therefore tend to grow faster in the following year, a short-run corrective force consistent with the state catch-up documented in earlier Brazilian studies (Canêdo-Pinheiro and Barbosa Filho, 2011; Schettini and Azzoni, 2018).

Long-run dispersion moves in the opposite direction. Between 1995 and 2023, sigma rises from 0.089 to 0.234 in the aggregate economy, from 0.640 to 1.321 in agriculture, from 0.144 to 0.494 in industry, and from 0.188 to 0.625 in commerce and services. Agriculture reaches the widest cross-state distribution by the end of the sample, while industry and commerce and services also move markedly

Figure 1. Sectoral TFP trajectories across Northeastern states, 1986–2023



Notes: Each state-sector index is normalized to 1986=100. The dotted horizontal line marks the within-state baseline and does not represent an absolute cross-state productivity level.

Source: Authors' calculations.

Table 2. Short-run catch-up, long-run dispersion, and five-year persistence

<i>Panel A. Dispersion and panel catch-up, 1995–2023</i>					
Sector	σ_{1995}	σ_{2023}	$\Delta\sigma$	$\hat{\beta}$	Wild p
Aggregate economy	0.089	0.234	0.146	-0.070	0.010
Agriculture	0.640	1.321	0.681	-0.115	0.010
Industry	0.144	0.494	0.349	-0.177	0.006
Commerce and services	0.188	0.625	0.437	-0.065	0.010
<i>Panel B. Five-year persistence in relative-productivity regimes</i>					
Sector	Low→Low	Middle→Middle	High→High	Transitions	
Aggregate economy	0.667	0.485	0.763	297	
Agriculture	0.879	0.644	0.862	297	
Industry	0.809	0.486	0.747	297	
Commerce and services	0.900	0.531	0.871	279	

Notes: σ is the cross-state standard deviation of log TFP. Panel A estimates Equation (6) with state and year fixed effects; reported p -values use wild-cluster inference at the state level. Panel B classifies states annually into low, middle, and high relative-productivity regimes and reports five-year diagonal transition probabilities.

Source: Authors' calculations.

away from their 1995 dispersion. Faster subsequent growth among lagging states is therefore too episodic or too unevenly distributed to close the cumulative gaps. The distinction emphasized by Quah (1993) appears clearly within one regional system: mean reversion in growth can coexist with an increasingly dispersed cross section.

Panel B shows the persistence underlying that coexistence. At a five-year horizon, the low and high regimes are much more persistent than the middle of the distribution. In agriculture, 87.9% of observations initially in the low regime remain low five years later and 86.2% of high-regime observations remain high. Industry records 80.9% and 74.7%, respectively. Commerce and services is even more persistent at the extremes, with 90.0% remaining low and 87.1% remaining high. The aggregate economy displays greater mobility, yet two-thirds of low observations and more than three-quarters of high observations still occupy the same regime after five years. Short-run catch-up consequently occurs inside a distribution whose endpoints are difficult to leave.

The Phillips–Sul tests reinforce this distributional reading. Global convergence is rejected for all four sectors; the log- t statistics are -7.86 for the aggregate economy, -11.55 for agriculture, -16.03 for industry, and -23.30 for commerce and services. The global tests rule out a common transition path and reveal structured heterogeneity in the subsequent clustering. Agriculture produces the clearest club hierarchy and is examined in the next subsection; Appendix Table A3 reports the classifications for all sectors.

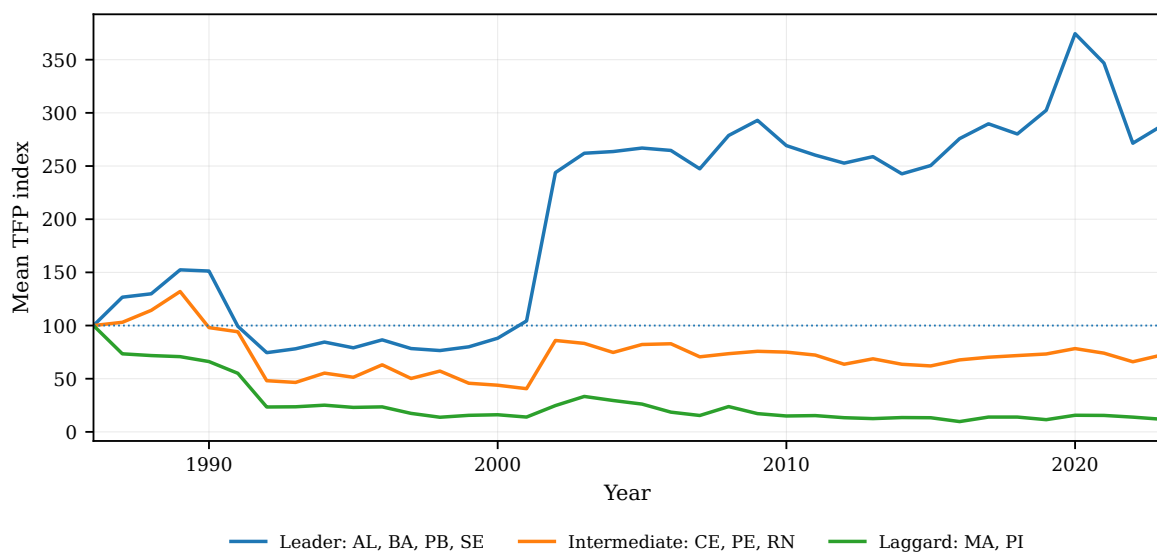
The pattern remains under alternative technology and sample choices. Fixed-coefficient OLS, FLS anchoring bands of 0.5 and 1.5 OLS standard errors, and FLS estimates with nonnegative input coefficients all preserve rising dispersion and negative beta coefficients. Windows beginning in 1987 and 2000 do the same, while leave-one-state-out estimates keep $\Delta\sigma > 0$ and $\beta < 0$ in every sector. Targeted measurement checks also leave the sign pattern unchanged when the constructed capital stock is omitted from the aggregate, agricultural, or industrial production equation, or when other sector-specific inputs are removed in turn. Appendix Tables A1 and A2 report these diagnostics. Catch-up without convergence is therefore a stable property of the estimated state-sector paths across materially different measurement choices.

Economically, the region exhibits correction without durable reordering. States recover after relative setbacks, but those gains rarely accumulate into sustained movement from the bottom to the top of the distribution. Convergence requires productivity improvements to diffuse with enough breadth and persistence to alter the hierarchy itself. Agriculture and industry show two distinct ways in which that diffusion remains incomplete.

4.3. Uneven agricultural productivity upgrading

Agriculture provides the sharpest example of productivity upgrading generating internal stratification. The Phillips–Sul procedure identifies three groups with distinct transition paths. Alagoas, Bahia, Paraíba, and Sergipe form the leading club; Ceará, Pernambuco, and Rio Grande do Norte form an intermediate group; Maranhão and Piauí constitute the lagging club. Figure 2 plots the annual mean trajectory of each group and shows that the paths separate well before the end of the sample.

Figure 2. Agricultural TFP trajectories by Phillips–Sul convergence club



Notes: Club trajectories are annual averages of state TFP indices within the formal Phillips–Sul agricultural clubs. All state indices equal 100 in 1986 by construction.

Source: Authors' calculations.

The magnitudes in Table 3 are economically large. In 1995, the leading club has an average TFP index of 79.1 and the lagging club 23.0. By 2023, the corresponding values are 287.9 and 11.9. In mean relative log TFP, the leader–laggard gap expands from about 1.12 to 3.14. The intermediate club moves from 51.4 to 72.3, advancing relative to its own 1986 benchmark but much less than the leading group. Agricultural productivity upgrading therefore takes the form of an increasingly differentiated regional hierarchy, with the largest gains concentrated in a subset of states.

The club structure is consistent with a broader literature in which Brazilian agricultural transformation is spatially heterogeneous and productive sophistication is associated with large regional differences (Stein et al., 2024; Antunes and Freitas, 2025). Here, that heterogeneity appears as persistent state-level TFP transition paths over nearly four decades. The broader divergence result is also insensitive to the construction of any single agricultural input: fixed-coefficient checks that omit physical capital, land, or human capital separately continue to yield rising dispersion and negative short-run correction coefficients (Appendix Table A2).

The MATOPIBA comparison further clarifies the regional scale of the result. Maranhão and Piauí contain important frontier areas, yet the TFP indices summarize each state's entire agricultural sector and measure cumulative change relative to its own 1986 benchmark. Frontier expansion through land and capital can therefore coexist with a weaker statewide residual TFP path. The cross-state divergence also extends beyond the three Northeastern states associated with MATOPIBA: after Bahia, Maranhão,

Table 3. Agricultural club trajectories and robustness of regional divergence

<i>Panel A. Formal Phillips–Sul agricultural clubs</i>				
Club	States	Mean TFP 1995	Mean TFP 2023	Change
Leader	AL, BA, PB, SE	79.1	287.9	208.8
Intermediate	CE, PE, RN	51.4	72.3	20.9
Laggard	MA, PI	23.0	11.9	-11.2
<i>Panel B. Agricultural dispersion with and without MATOPIBA-related states</i>				
Sample	States excluded	σ_{1995}	σ_{2023}	$\Delta\sigma$
All North-eastern states	None	0.640	1.321	0.681
Excluding MATOPIBA-related states	BA, MA, PI	0.491	0.856	0.365

Notes: The Phillips–Sul global log- t statistic for agriculture is -11.55 , rejecting a common transition path. Mean TFP indices remain normalized to each state’s own 1986 value. Panel B removes Bahia, Maranhão, and Piauí jointly; MATOPIBA covers only parts of these states, so the exclusion is geographically broader than the frontier itself.

Source: Authors’ calculations.

and Piauí are excluded jointly, agricultural sigma rises from 0.491 in 1995 to 0.856 in 2023. The widening agricultural hierarchy is consequently a broader Northeastern phenomenon.

Agriculture illustrates why rapid productivity upgrading need not produce regional homogenization. Productivity gains are strong enough to transform the leading states, but their diffusion is too spatially concentrated to draw the regional distribution toward a common path. In this setting, the pace of upgrading and its geographic reach are separate dimensions of development.

4.4. Industrial divergence within comparable activities

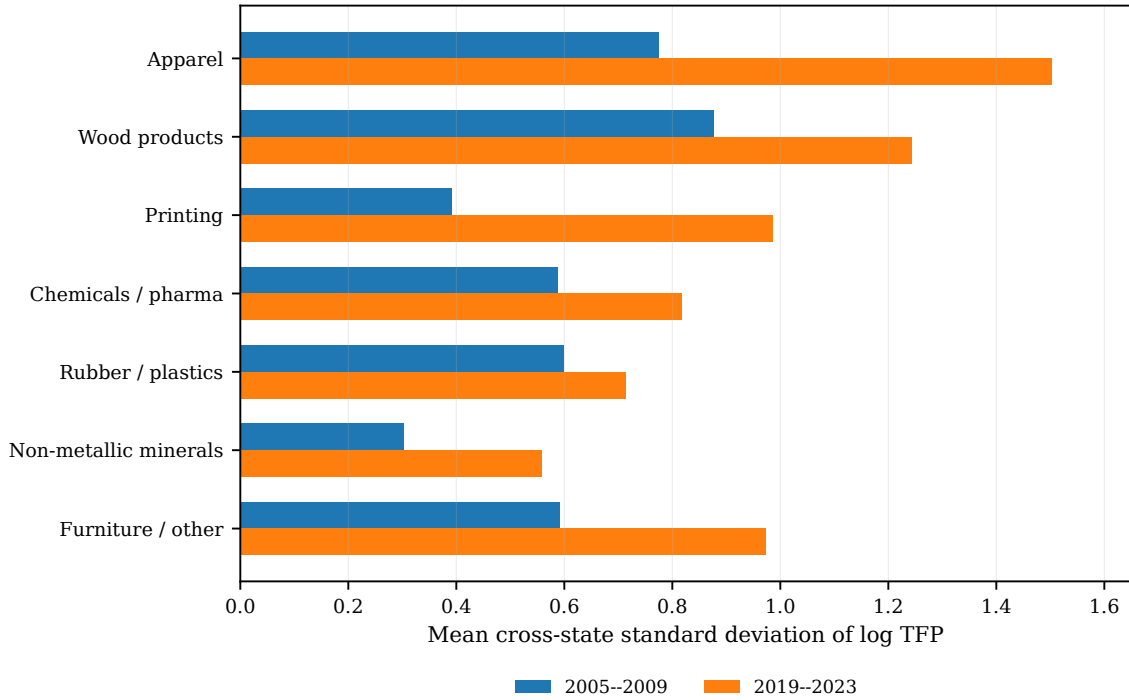
To separate location-specific productivity dynamics from differences in industrial composition, we compare the nine states within seven manufacturing activities that are observed continuously everywhere from 1998 to 2023. This balanced design holds broad activity constant and asks whether the same types of industry converge across locations.

Figure 3 shows that cross-state dispersion increases in all seven activities. Apparel records the largest widening, with mean sigma rising from 0.774 in 2005–2009 to 1.503 in 2019–2023. Printing moves from 0.391 to 0.987, wood products from 0.877 to 1.244, and furniture and related activities from 0.592 to 0.973. Even the smallest increase, in rubber and plastics, is positive. Averaging across the seven activities, cross-state dispersion rises from 0.589 to 0.971, approximately 65%.

The breadth of the pattern is economically more revealing than any single activity. Every balanced industry becomes more geographically dispersed even though the underlying activity is held fixed. State industrial divergence therefore reflects more than a changing mix of sectors: comparable industries themselves generate increasingly different productivity trajectories across locations. The same conclusion appears with fixed-coefficient cell TFP, for which all seven activities again widen, and across alternative early comparison windows. Appendix Table A4 reports these checks.

Medium-run mobility inside manufacturing is similarly limited. Across the 63 balanced state–industry cells, 78.9% of observations in the low-productivity regime remain low five years later, while 72.8% of high-regime observations remain high. Combined with the state-level industrial paths, these transitions describe a coherent pattern: productivity differences widen within comparable activities and relative positions remain persistent over economically meaningful horizons.

Figure 3. Cross-state TFP dispersion within balanced manufacturing industries



Notes: The sample contains seven manufacturing activities observed continuously in all nine states. Dispersion is computed across states within each activity using cell-level FLS TFP.

Source: Authors' calculations from the balanced PIA–RAIS manufacturing panel.

Regional production research provides a natural interpretation for this within-industry evidence. Manufacturing agglomeration in Brazil is highly localized (Almeida et al., 2023), while diversification into new activities depends strongly on capabilities related to what a region already knows how to produce (Françoso et al., 2024; Freitas et al., 2024). Recent evidence also documents skill upgrading in less industrialized Brazilian regions alongside significant spatial spillovers (Castro and Azzoni, 2025). These findings imply that a common activity classification can coexist with distinct local productive environments. Our results show that location remains economically relevant to the productivity paths of comparable manufacturing activities.

Agriculture and industry thus reproduce the regional hierarchy through different sectoral processes. Agricultural upgrading is substantial but spatially concentrated, creating clearly separated clubs. Industrial efficiency is weak more broadly and diverges even within the same activities. These patterns explain the central tension in Table 2: short-run corrective growth is present, but sectoral transformation does not diffuse uniformly enough to overturn persistent regional positions.

5. Conclusion

Short-run catch-up has not eroded Northeastern Brazil's regional productivity hierarchy. Across the aggregate economy, agriculture, industry, and commerce and services, states with lower relative TFP grow faster in the following year, yet cross-state dispersion increases from 1995 to 2023. Five-year transitions complete the picture: low- and high-productivity positions remain substantially more persistent than the middle of the distribution. The region therefore combines a recurrent corrective force with widening cumulative productivity paths.

Sectoral evidence shows how this pattern is formed. Agriculture exhibits the largest productivity advances and, at the same time, the strongest internal stratification. A leading group accelerates sharply, an intermediate group progresses more slowly, and Maranhão and Piauí follow much weaker statewide trajectories; the increase in agricultural dispersion remains substantial after the Northeastern MATOPIBA states are excluded jointly. Industry displays a different process. State-level TFP paths are

weak across most of the region, and cross-state dispersion rises in every one of the seven manufacturing activities observed continuously across all nine states. Holding broad activity constant therefore leaves a pronounced and persistent geographic separation in productivity paths.

These results shift the regional convergence question from temporary growth differentials to the diffusion of productivity upgrading. A lagging macro-region can experience structural transformation and repeated episodes of catch-up while becoming more internally differentiated. In agriculture, productivity upgrading is distributed unevenly across space; in industry, comparable activities follow distinct productivity paths across locations. Within-sector dynamics are consequently central to the persistence of regional disparities and help explain why aggregate indicators can understate the degree of productive heterogeneity inside the Northeast.

For regional policy, this distinction changes the relevant benchmark. Development strategies should evaluate whether factor accumulation and sectoral expansion translate into productivity gains that spread beyond already successful locations. In agriculture, the relevant margin is the geographic diffusion of upgrading beyond the leading productivity club. In industry, investment and output expansion should be distinguished from whether comparable activities convert accumulated inputs into sustained efficiency growth across locations. The decisive benchmark is therefore not whether lagging places occasionally grow faster, but whether productivity gains become broad and persistent enough to change the regional distribution.

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Appendix

The appendix consolidates the estimator, sample, distributional, and within-industry checks referenced in the main text. Each exercise preserves the empirical distinction between short-run corrective growth and widening cumulative productivity paths.

Table A1. TFP estimator and FLS sensitivity

Sector	Baseline FLS		Fixed OLS		Nonnegative FLS	
	$\Delta\sigma$	$\widehat{\beta}$	$\Delta\sigma$	$\widehat{\beta}$	$\Delta\sigma$	$\widehat{\beta}$
Aggregate economy	0.146	-0.070	0.108	-0.131	0.083	-0.122
Agriculture	0.681	-0.115	0.684	-0.127	0.684	-0.126
Industry	0.349	-0.177	0.271	-0.248	0.251	-0.256
Commerce and services	0.437	-0.065	0.522	-0.049	0.169	-0.139
<i>Alternative FLS anchoring bands</i>						
Sector	0.5 OLS s.e.		1.5 OLS s.e.			
	$\Delta\sigma$	$\widehat{\beta}$	$\Delta\sigma$	$\widehat{\beta}$		
Aggregate economy	0.112	-0.098	0.183	-0.048		
Agriculture	0.658	-0.113	1.093	-0.082		
Industry	0.331	-0.188	0.374	-0.124		
Commerce and services	0.430	-0.068	0.425	-0.063		

Notes: Diagnostics use the 1995–2023 convergence window. Fixed OLS holds input coefficients constant within each state-sector series. Nonnegative FLS imposes nonnegative slope coefficients on observed inputs. The baseline anchoring band is one Newey–West OLS standard error; the lower panel varies that band while retaining the same selection criterion.

Source: Authors' calculations.

Table A2. Alternative convergence windows and sample robustness

<i>Panel A. Alternative convergence windows</i>				
	1987–2023		2000–2023	
Sector	$\Delta\sigma$	$\widehat{\beta}$	$\Delta\sigma$	$\widehat{\beta}$
Aggregate economy	0.172	-0.081	0.143	-0.111
Agriculture	1.024	-0.109	0.494	-0.235
Industry	0.410	-0.122	0.301	-0.234
Commerce and services	0.580	-0.046	0.375	-0.069
<i>Panel B. Leave-one-state-out ranges, 1995–2023</i>				
Sector	$\Delta\sigma$ range	$\widehat{\beta}$ range		
Aggregate economy	[0.047, 0.173]	[-0.123, -0.074]		
Agriculture	[0.543, 0.835]	[-0.173, -0.120]		
Industry	[0.218, 0.374]	[-0.274, -0.151]		
Commerce and services	[0.284, 0.483]	[-0.081, -0.043]		
<i>Panel C. Targeted fixed-coefficient measurement checks, 1995–2023</i>				
Sector	Specification	$\Delta\sigma$	$\widehat{\beta}$	
Aggregate economy	Excluding physical capital	0.226	-0.070	
Agriculture	Excluding physical capital	0.516	-0.153	
Agriculture	Excluding land	0.591	-0.140	
Agriculture	Excluding human capital	0.685	-0.130	
Industry	Excluding physical capital	0.584	-0.143	
Industry	Excluding human capital	0.106	-0.307	
Commerce and services	Excluding electricity proxy	0.687	-0.044	

Notes: The main window in Table 2 is 1995–2023. Leave-one-state-out columns report the minimum and maximum statistic after excluding each state in turn. Measurement checks use fixed-coefficient OLS and remove the listed input from the production equation. These omission exercises are deliberately stringent tests of whether the convergence pattern depends on a particular constructed or sector-specific factor measure.

Source: Authors' calculations.

Table A3. Phillips–Sul club structure and additional mobility evidence

Sector	Global log- t	Formal convergence clubs	States outside formal clubs
Aggregate economy	-7.86	AL, BA, CE, PE, PI, SE	MA, PB, RN
Agriculture	-11.55	AL, BA, PB, SE; CE, PE, RN; MA, PI	–
Industry	-16.03	MA, PB, PE, PI, RN, SE	AL, BA, CE
Commerce and services	-23.30	PI, SE; MA, PB	AL, BA, CE, PE, RN

Notes: Negative global log- t statistics reject a common transition path under the Phillips–Sul procedure. Semicolons separate formal clubs returned by the clustering algorithm. Agricultural club ordering follows terminal relative TFP and is reported in Table 3.

Five-year mobility beyond the diagonal probabilities in Table 2

Sector	Up from low	Down from high	Transitions
Aggregate economy	0.333	0.237	297
Agriculture	0.121	0.138	297
Industry	0.191	0.253	297
Commerce and services	0.100	0.129	279

Notes: Regimes use pooled sector-specific terciles of annual relative log TFP. “Up from low” is the probability of leaving the low regime within five years; “Down from high” is the probability of leaving the high regime.

Source: Authors’ calculations.

Table A4. Within-industry divergence: estimator, window, and mobility checks

Specification	Rising activities	Early σ	Late σ	Mean change
Baseline FLS, 2005–09 vs. 2019–23	7/7	0.589	0.971	0.382
Fixed OLS, 2005–09 vs. 2019–23	7/7	0.555	0.891	0.336
FLS, 2003–07 vs. 2019–23	7/7	0.495	0.971	0.476
FLS, 2004–08 vs. 2019–23	7/7	0.544	0.971	0.426
FLS, 2006–10 vs. 2019–23	7/7	0.641	0.971	0.330
FLS, 2008–12 vs. 2019–23	7/7	0.715	0.971	0.256
FLS, 2005–09 vs. 2018–22	7/7	0.589	0.934	0.345

Notes: Early and late sigma are averages of annual cross-state dispersion computed within each balanced manufacturing activity. Alternative rows vary the cell-level TFP estimator or the comparison window.

Five-year transitions across the 63 balanced state–industry cells

	Low→Low	Middle→Middle	High→High	Transitions
Balanced manufacturing cells	0.789	0.592	0.728	882

Notes: Transition regimes are defined from the relative cell-level TFP distribution.

Source: Authors’ calculations from the balanced PIA–RAIS manufacturing panel.